

# MANISH KUMAR SABBANI

## Sr. Data Analyst

Phone: +1 469-630-1694

Email: smanishk495@gmail.com

LinkedIn URL: [www.linkedin.com/in/manish-kumar-s](https://www.linkedin.com/in/manish-kumar-s)



### **PROFESSIONAL SUMMARY:**

*Dynamic and detail-oriented Senior Data Analyst with 10+ years of IT experience, including 6+ years working with Azure Data Services, Big Data ecosystems, and cloud platforms. Skilled at designing data models, performing in-depth analyses, and delivering actionable insights that drive data-informed decision-making. Adept at blending strong analytical expertise, SQL optimization, and BI visualization skills with a deep understanding of data architecture and cloud-based solutions. Proven ability to collaborate with crossfunctional teams and translate complex data into clear, meaningful business outcomes.*

- Over all 10 years of professional IT experience and over 6 years of Azure with Big Data Ecosystem experience in ingestion, storage, querying, processing and analysis of big data/ Azure Data bricks/cloud Technologies and 4 years' experience in Data warehouse and ETL developer.
- Experienced data professional with a strong background in end-to-end management of ETL data pipelines, ensuring scalability and smooth operations.
- Designed and implemented scalable ETL/ELT pipelines using Azure Data Factory, Databricks, Apache Spark (Spark SQL, Spark Data Frames, Spark Core), Apache Kafka, Apache Airflow, Apache Beam, and Snowflake.
- Strong SQL expertise (DDL, DML) with experience in indexing, query optimization, and advanced database objects for performance tuning.
- Proficient in data warehousing concepts, including data cleansing, Slowly Changing Dimensions (SCD), surrogate key creation, and Change Data Capture (CDC) for Snowflake modelling
- Integrated data from on-premises and cloud sources into Snowflake and other targets using ADF, Glue, and custom ingestion frameworks.
- Developed streaming data pipelines using Kafka, Spark Streaming, Apache Hive, and Apache HBase for real-time ingestion, transformation, and analytics.
- Built and optimized data models and schemas in Snowflake, Hive, and HBase for analytics and reporting workloads.
- Applied AI/ML frameworks such as PySpark MLlib, TensorFlow, PyTorch, and Pandas for large-scale predictive modelling, feature engineering, and automated machine learning workflows in Azure Databricks and AWS EMR.
- Integrated ML model training, validation, and deployment into Airflow and ADF pipelines for end-to-end production ML solutions on cloud data platforms.
- Skilled in executing Hive scripts via Hive on Spark, leveraging partitioning, bucketing, and compression for performance optimization.
- Created ETL transformations and validations using Spark-SQL/Spark Data Frames within Azure Databricks and AWS Glue.
- Applied data quality checks, validation frameworks, and cleansing techniques to ensure accuracy, consistency, and governance.
- Automated data workflows and integrations using Azure Logic Apps, Azure Functions, AWS Lambda, and AWS Step Functions
- Implemented CI/CD pipelines for data engineering solutions using Jenkins, enabling automated deployment and testing.
- Leveraged AWS S3 for raw and curated data storage, AWS Redshift & Athena for analytics, and AWS Lake Formation for centralized governance and access control.
- Designed both batch and micro-batch workflows using Apache Beam, Spark, and Kinesis Data Streams for high-volume data processing.
- Integrated curated datasets with BI tools including Amazon Quick Sight, Power BI, and Tableau for interactive dashboards and reporting.
- Extensive hands-on experience with Dataiku for data preparation, machine learning, model deployment, and building end-to-end data science workflows in collaborative environments.
- Utilized JIRA for Agile project management, sprint planning, daily stand-ups, and progress tracking.
- Actively participated in Agile ceremonies, collaborating cross-functionally with data engineers, DevOps, and BI teams to ensure delivery of high-quality solutions
- Delivered end-to-end multi-cloud data engineering solutions ensuring scalability, performance, and high availability across Azure and AWS platforms.

- Continuously evaluated and adopted emerging technologies and tools to enhance data pipeline efficiency, scalability, and maintainability in a fast-evolving cloud environment.

## TECHNICAL SKILLS

|                                  |                                                                                                                                                                         |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Azure Services                   | Azure data Factory, Airflow, Azure Data Bricks, Logic Apps, Functional App, Snowflake, Azure DevOps, Azure Synapse Analytics, Microsoft Fabric, Azure Machine Learning. |
| AWS Services                     | S3, Glue, EMR, Redshift, Athena, Kinesis, Lambda, Step Functions, Lake Formation, Quick Sight, SageMaker.                                                               |
| Big Data & AI/ML Technologies    | MapReduce, Hive, Dataiku, Python, PySpark, PySpark MLlib, Scala, Kafka, Spark streaming, Oozie, Sqoop, Zookeeper, TensorFlow, PyTorch, Keras.                           |
| Data Platforms/ Enterprise Tools | Palantir, Dataiku, Microsoft Fabric, Snowflake, Azure Synapse, Databricks.                                                                                              |
| Data Visualization               | Power BI, Tableau, Amazon Quick Sight.                                                                                                                                  |
| Hadoop Distribution              | Cloudera, Horton Works.                                                                                                                                                 |
| Languages                        | Java, SQL, PL/SQL, Python, HiveQL, Scala.                                                                                                                               |
| Operating Systems                | Windows (XP/7/8/10), UNIX, LINUX, UBUNTU, CENTOS.                                                                                                                       |
| Build Automation tools           | Ant, Maven                                                                                                                                                              |
| Version Control                  | GIT, GitHub.                                                                                                                                                            |
| IDE & Build Tools, Design        | Eclipse, Visual Studio.                                                                                                                                                 |
| Databases                        | MS SQL Server 2016/2014/2012, Azure SQL DB, Azure Synapse. MS Excel, MS Access, Oracle 11g/12c, Cosmos DB                                                               |

## EDUCATION

- Bachelors in Electronics & Communication Engineering from Jawaharlal Nehru Technological University Hyderabad, India.
- Masters in Information Technology at University of North Texas

## CERTIFICATION

- AWS Certified Data Engineer – Associate.
- Microsoft Certified - Azure Solution Architect.

## WORK EXPERIENCE

**Role: Sr. Data Analyst**

**Feb 2024 – Till Now**

**Client: FM Global, Johnston, RI Responsibilities:**

- Performed end-to-end data analysis, preparation, and transformation using Dataiku and Azure Data Factory (ADF), ensuring data accuracy, consistency, and readiness for reporting and analytics.
- Performed end-to-end data analysis, preparation, and transformation using Palantir Foundry, Dataiku, and Azure Data Factory (ADF), ensuring data accuracy, consistency, and readiness for reporting and analytics.
- Leveraged Palantir pipelines, Dataiku visual recipes, SQL, and Python to clean, enrich, and transform datasets for exploratory analysis and business reporting.
- Designed and optimized SQL queries, indexes, and views in Azure SQL Database, Snowflake, and Palantir Foundry to support fast and efficient analytical queries.
- Conducted in-depth analysis of on-premises and cloud data sources (MySQL, Cassandra, Azure Blob, Azure SQL DB, Snowflake, Palantir Foundry datasets) to derive actionable insights for business stakeholders.
- Applied data warehousing concepts (data cleansing, Slowly Changing Dimensions, surrogate keys, Change Data Capture) to support accurate reporting in Snowflake and Palantir Foundry.
- Built Power BI and Palantir dashboards and interactive reports by integrating curated datasets from Foundry, Dataiku, Databricks, and Snowflake, enabling real-time decision-making.
- Applied statistical and predictive analytics using Scikit-learn, TensorFlow, PyTorch, Pandas, and Palantir AI/ML modules within Foundry, Dataiku, and Azure Databricks, delivering insights for anomaly detection, forecasting, and classification tasks.

- Partnered with business teams to define KPIs, metrics, and data requirements, ensuring alignment between analytical outputs and organizational goals.
- Conducted data quality checks, profiling, and validation within Palantir Foundry, Dataiku, and Databricks, ensuring integrity and governance of analytical datasets.
- Automated recurring reports, workflows, and alerts using Azure Logic Apps, Azure Functions, Dataiku scenarios, and Palantir workflows, reducing manual effort and improving timeliness.
- Collaborated with cross-functional teams in Agile sprints using JIRA, contributing to requirement gathering, sprint planning, and progress tracking.
- Delivered ad-hoc analysis and deep dives into structured and unstructured datasets (SQL, Hive, HBase, ADLS, Foundry datasets) to support risk assessment, operational efficiency, and business strategy.
- Partnered with DevOps to version-control SQL scripts, analytical models, and dashboards using Git, Jenkins, and Azure DevOps for reliable deployments.
- Presented findings and recommendations through visual storytelling, Palantir dashboards, Power BI reports, and executive-level presentations.

**Environment:** Palantir Foundry, Azure Databricks, Dataiku, Data Factory, Logic Apps, Function Apps, Snowflake, MS SQL, Oracle, HDFS, Hive, Python, SQL, PySpark, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, PyTorch, Power BI, Git, JIRA, Jenkins, Kafka.

**Role: Sr. Data Analyst**

**Nov 2021 – Dec 2023**

**Client: Discover Financial, Riverwoods, IL Responsibilities:**

- Analyzed large datasets ingested from databases, APIs, and file systems using Dataiku, Azure Data Factory (ADF), and Kafka, ensuring data accuracy and business readiness.
- Worked as a **Data Analyst** analyzing large datasets ingested from databases, APIs, and file systems using Dataiku, Azure Data Factory (ADF), and Kafka, ensuring data accuracy and business readiness.
- Used Dataiku visual recipes, SQL, and Python to clean, transform, and enrich datasets for reporting and analytics purposes.
- Designed and optimized SQL queries, indexes, and stored procedures in Azure SQL Database, Snowflake, and Oracle to improve reporting and dashboard performance.
- Migrated analytical datasets from on-premises Oracle to Azure Synapse Analytics and Data Lake, enabling cloud-based analytics and BI reporting.
- Built and maintained Power BI dashboards by integrating curated datasets from Dataiku and Azure Databricks, supporting risk, compliance, and customer analytics teams.
- Applied statistical analysis and predictive modeling using Dataiku AutoML, Scikit-learn, TensorFlow, PyTorch, Spark MLlib, and Pandas to generate actionable business insights.
- Partnered with business stakeholders to define KPIs, metrics, and reporting requirements, delivering insights aligned with organizational goals.
- Conducted data quality checks, profiling, and validation in Dataiku and Databricks to ensure trusted and consistent datasets.
- Delivered batch and streaming analytics using Dataiku streaming plugins, Spark Streaming, and Kafka, enabling near real-time reporting and dashboards.
- Created data models and schemas in Snowflake, Hive, and Parquet to support efficient BI, governance, and self-service analytics.
- Automated recurring reporting workflows and alerts using Azure Logic Apps and Functions, improving timeliness and reducing manual effort.
- Collaborated in Agile sprints using JIRA, contributing to sprint planning, issue tracking, and requirement gathering.
- Presented insights and recommendations through visual storytelling, Power BI reports, and executive-level presentations.

**Environment:** Azure Databricks, Dataiku, Data Factory, Logic Apps, Function Apps, Snowflake, MS SQL, Oracle, HDFS, Hive, Python, SQL, PySpark, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, PyTorch, Power BI, Git, JIRA, Jenkins, Kafka.

**Role: Data Analyst**

**Mar 2019 – Sept 2021**

**Client: State of Oregon(Dept. Of Health), Portland, OR**

**Responsibilities:**

- Designed and supported an **Enterprise Data Lake** on **Azure Data Lake Storage (ADLS)** to enable analytics, reporting, and storage of large-scale healthcare datasets.

- Performed **data cleaning, transformation, and validation** across relational, cloud, and semi-structured environments to ensure accuracy, completeness, and consistency of sensitive healthcare data.
- Built **tabular and semantic models in Azure Analysis Services** to support advanced business reporting, self-service analytics, and executive-level dashboards.
- Extracted, ingested, and prepared data from multiple sources including **on-premises SQL/Oracle databases, Azure Blob, ADLS, HDFS, and HBase** using Azure Data Factory (ADF), PySpark, and SQL, ensuring seamless integration into analytics and BI platforms.
- Developed **interactive dashboards and analytical datasets in Tableau and SSRS**, providing actionable insights into healthcare operations, patient outcomes, compliance metrics, and cost-performance trends.
- Created and optimized **end-to-end data pipelines in Azure (ADF + Databricks)** for automated data preparation, cleansing, and aggregation, improving accuracy and reducing manual intervention for reporting and visualization.
- Applied **advanced SQL, Hive queries, and Spark SQL** for data aggregation, KPI development, cohort analysis, trend identification, and predictive insights supporting healthcare policy and operational decisions.
- Conducted **data profiling, quality assurance, and reconciliation** across datasets to ensure reliability and consistency for analysts, leadership, and regulatory reporting requirements.
- Performed **ad-hoc data analysis using Python (Pandas, PySpark)** to identify anomalies, detect trends, optimize resource utilization, and provide actionable recommendations for clinical and operational decision-making.
- Partnered closely with healthcare business teams, stakeholders, and data stewards to define **KPIs, metrics, and reporting requirements**, translating complex data into clear, impactful visual insights.
- Automated **recurring reporting workflows and data refresh cycles** using Databricks notebooks, Python scripts, and ADF pipelines, significantly improving reporting efficiency and reducing turnaround time for insights.
- Developed and maintained **compliance-focused data documentation and metadata**, ensuring traceability and audit readiness for regulatory reviews.
- Presented actionable insights through **executive-level dashboards, PowerPoint presentations, and visual storytelling**, enabling leadership to make informed decisions on patient care, operations, and strategic initiatives.
- Conducted **root cause analysis and trend analysis** for operational inefficiencies, patient outcomes, and cost management, providing strategic recommendations to improve healthcare delivery.
- Collaborated in **cross-functional Agile teams**, participating in sprint planning, requirement gathering, and delivery of high-quality analytics solutions within healthcare programs.

**Environment:** Azure Data Lake Storage (ADLS), Azure Data Factory, Azure SQL Database, Azure Synapse Analytics, Azure Databricks, Apache Spark, PySpark, Python, Hive, SQL, SSRS, Tableau, Power BI, Snowflake, Git.

## **Role: Big data Developer**

**Jun 2018 – Jan 2020**

### **Client: AT&T, Dallas, TX Responsibilities:**

- Designed and developed scalable data transformation applications on AWS Data Lake to enable efficient analytics for business stakeholders.
- Built and optimized data ingestion pipelines using AWS Glue to ingest data from diverse sources into Amazon S3, Amazon RDS, and Amazon Redshift.
- Developed complex data flows and transformations leveraging PySpark within AWS EMR (Elastic MapReduce), enhancing data processing performance and scalability.
- Implemented batch and streaming data processing solutions using Apache Spark and Spark Streaming on EMR to handle real-time analytics requirements.
- Created and managed schema designs and data models in Amazon Redshift and Snowflake to support fast and reliable reporting and BI.
- Automated data workflows using AWS Step Functions and AWS Lambda for event-driven and scheduled data processing.
- Leveraged AWS Code Pipeline and Jenkins to establish CI/CD pipelines, ensuring smooth deployment of data engineering solutions.
- Integrated big data tools such as Apache Kafka, Hive, HBase, and Sqoop with AWS services to facilitate large-scale data ingestion and processing.
- Utilized Shell scripting for automating ETL workflows, managing system operations, and supporting data pipeline orchestration.
- Ensured data quality and governance by implementing validation checks and cleansing frameworks across pipelines.
- Collaborated with cross-functional teams to design end-to-end data architectures and meet evolving business needs.

**Environment:** AWS Cloud Services (S3, Glue, EMR, Redshift, Step Functions, Lambda, Code Pipeline), Cloudera CDH 3/4, Hadoop, HDFS, MapReduce, Hive, Oozie, Pig, Shell Scripting, MySQL, Apache Kafka, HBase, Sqoop, Jenkins.

**Role: Data warehouse Developer****Apr 2016 – Apr 2018****Client: Wells Fargo, Charlotte, NC Responsibilities:**

- Designed, created, and maintained databases for Server Inventory and Performance Inventory to support enterprise reporting requirements.
- Participated in Agile Scrum ceremonies; utilized Visual SourceSafe for version control and Trello for project tracking to ensure timely delivery.
- Developed drill-through and drill-down interactive reports with dynamic filters, sorting, and subtotal features using Power BI.
- Built Data Marts feeding downstream reports and developed a User Access Tool enabling users to create ad-hoc reports and query OLAP cubes.
- Created robust ETL packages using SSIS to extract data from heterogeneous sources, transform it, and load into the data warehouse.
  - Automated report generation and cube refresh processes through SSIS jobs, improving data availability and workflow efficiency.
- Deployed SSIS packages to production, employing environment-independent configurations for flexibility across deployment stages.
- Utilized SQL Server Reporting Services (SSRS) to design, author, and deliver both interactive web-based and paper-based reports.
- Developed complex stored procedures and triggers to ensure consistent and accurate data entry within the databases.
- Facilitated secure external data sharing using Snowflake, enabling rapid data exchange without building additional pipelines.

**Environment:** Windows server, MS SQL Server 2014, SSIS, SSAS, SSRS, SQL Profiler, Power BI, C#, Performance Point Server, MS Office, SharePoint

**Role: Data warehouse Developer****Mar 2014 – Feb 2016****Client: United Airlines, Chicago, IL Responsibilities:**

- Experience in developing complex store procedures, efficient triggers, required functions, creating indexes and indexed views for performance.
- Developed complex stored procedures, triggers, functions, indexes, and indexed views to optimize SQL Server performance.
- Monitored and tuned SQL Server environments for improved query execution and system efficiency.
- Designed and implemented ETL data flows using SSIS, creating mappings and workflows to extract and transform data from SQL Server, Access, and Excel sources.
- Performed dimensional data modelling for Data Mart design, including fact and dimension tables, and managed Slowly Changing Dimensions (SCD).
- Managed error and event handling in ETL processes using precedence constraints, breakpoints, checkpoints, and logging.
- Built and deployed SSAS cubes and dimensions with various architectures; authored MDX scripts to enhance BI functionality.
- Applied star and snowflake schema designs for efficient data warehousing solutions.
- Created parameterized, ad hoc, dashboard, and scorecard reports in SSRS, utilizing drill-down, drill-through, and cascading features.
- Collaborated effectively within teams, demonstrating strong communication and leadership skills to deliver client-focused solutions.

**Environment:** MS SQL Server 2016, Visual Studio 2017/2019, SSIS, SharePoint, MS Access, Team Foundation Server, Git.